

# Find the area of parallelograms

Use the scaffold, show each step and ask for an adult check when marked.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Read the model: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm. Copy or complete the first step.

2. Find the area of a triangle with base 15 cm and perpendicular height 8 cm.

3. A triangle has area 54 cm<sup>2</sup> and base 12 cm. Find its perpendicular height.

4. Miro says: Miro uses the sloping side as the height and forgets to halve a triangle's area. Is this correct? Explain.

## Teacher answers and checking prompts

1. Read the model: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm. Copy or complete the first step.  
**Answer** 84 cm<sup>2</sup>.
2. Find the area of a triangle with base 15 cm and perpendicular height 8 cm.  
**Answer** 60 cm<sup>2</sup>.
3. A triangle has area 54 cm<sup>2</sup> and base 12 cm. Find its perpendicular height.  
**Answer** 9 cm.
4. Miro says: Miro uses the sloping side as the height and forgets to halve a triangle's area. Is this correct? Explain.  
**Answer** Mark the perpendicular height, use base times height, and divide by two only for a triangle.

# Find the area of parallelograms

Work independently. Show a complete method and check at least one answer.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.

6. Check this result using an inverse, estimate or known fact: A triangle has area 54 cm<sup>2</sup> and base 12 cm. Find its perpendicular height.

2. Find the area of a triangle with base 15 cm and perpendicular height 8 cm.

7. Prove or explain your reasoning: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.

3. A triangle has area 54 cm<sup>2</sup> and base 12 cm. Find its perpendicular height.

8. Identify and correct this misconception: Miro uses the sloping side as the height and forgets to halve a triangle's area.

4. Solve this independently and show every stage: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Find the area of a triangle with base 15 cm and perpendicular height 8 cm.

10. Write and solve a similar question to: A triangle has area 54 cm<sup>2</sup> and base 12 cm. Find its perpendicular height.

# Teacher answers and checking prompts

1. Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.  
**Answer** 84 cm<sup>2</sup>.
2. Find the area of a triangle with base 15 cm and perpendicular height 8 cm.  
**Answer** 60 cm<sup>2</sup>.
3. A triangle has area 54 cm<sup>2</sup> and base 12 cm. Find its perpendicular height.  
**Answer** 9 cm.
4. Solve this independently and show every stage: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.  
**Answer** 84 cm<sup>2</sup>.
5. Use a second method or representation for: Find the area of a triangle with base 15 cm and perpendicular height 8 cm.  
**Answer** Expected result: 60 cm<sup>2</sup>. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A triangle has area 54 cm<sup>2</sup> and base 12 cm. Find its perpendicular height.  
**Answer** Expected result: 9 cm. A valid check must be shown.
7. Prove or explain your reasoning: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.  
**Answer** 80 cm<sup>2</sup>. The explanation must show why the method works.
8. Identify and correct this misconception: Miro uses the sloping side as the height and forgets to halve a triangle's area.  
**Answer** Mark the perpendicular height, use base times height, and divide by two only for a triangle.
9. Write a complete sentence explaining how you checked one answer.  
**Answer** Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A triangle has area 54 cm<sup>2</sup> and base 12 cm. Find its perpendicular height.  
**Answer** Answers vary. The new question must use the same mathematical structure and include a correct solution.

# Find the area of parallelograms

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Prove or explain your reasoning: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.

6. Create a new example that tests the same idea as: A triangle has area 54 cm<sup>2</sup> and base 12 cm. Find its perpendicular height.

2. Prove the answer to this question, rather than only calculating it: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Find the area of a triangle with base 15 cm and perpendicular height 8 cm.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 9 cm.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro uses the sloping side as the height and forgets to halve a triangle's area.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

# Teacher answers and checking prompts

1. Prove or explain your reasoning: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.  
**Answer** 80 cm<sup>2</sup>. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.  
**Answer** Expected result: 84 cm<sup>2</sup>. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Find the area of a triangle with base 15 cm and perpendicular height 8 cm.  
**Answer** Expected result: 60 cm<sup>2</sup>. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 9 cm.  
**Answer** One valid reconstruction is: A triangle has area 54 cm<sup>2</sup> and base 12 cm. Find its perpendicular height. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro uses the sloping side as the height and forgets to halve a triangle's area.  
**Answer** Mark the perpendicular height, use base times height, and divide by two only for a triangle.
6. Create a new example that tests the same idea as: A triangle has area 54 cm<sup>2</sup> and base 12 cm. Find its perpendicular height.  
**Answer** Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.  
**Answer** Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.  
**Answer** Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.  
**Answer** Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.  
**Answer** Answers vary. The explanation must distinguish the mathematical structure from the changed value.